Rigorous Design Notes and Code Guide for Improving the Conditional WGAN Synthetic-Genotype Pipeline

Technical companion for `conditional_wgan_pipeline.py`

October 2026

Abstract

This document explains and formalizes a set of opt-in improvements to a conditional Wasserstein GAN with gradient penalty (cWGAN-GP) for synthetic genotype generation. It is written for a reader who is relatively new to population genetics and generative modeling. Every proposed change is defined mathematically, the failure mode it targets is stated precisely, and the corresponding implementation is identified. The revised code keeps the original PCA-space MLP as the default baseline and adds experimental alternatives rather than silently changing the scientific model. The main additions are: (i) a direct SNP-space residual 1-D convolutional WGAN; (ii) FiLM conditioning at every hidden block; (iii) optional external ancestry coefficients such as ADMIXTURE estimates; (iv) inverse-frequency population-balanced minibatch sampling; (v) PacGAN-style critic packing; (vi) multi-metric model/run selection; and (vii) separate privacy diagnostics. The discussion also explains important limitations, including the distinction between unphased diploid genotype dosage and phased haplotypes, the effect of PCA truncation on covariance, and why privacy diagnostics are not differential-privacy guarantees.

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1 Code organization and reproducible experiment strategy

The implementation intentionally preserves the latest baseline pipeline and makes every new modeling idea an explicit switch. This is important scientifically: if architecture, data weighting, conditioning, and output representation are changed simultaneously, an improvement or degradation cannot be attributed to any one cause.

1.1 Files

File	Purpose
`conditional_wgan_pipeline.py`	Main run orchestration, data loading, PCA evaluation, logging, plotting, and CLI. The default backend remains the previous PCA-space model.
`cwgan_extensions/models.py`	FiLM-conditioned PCA WGAN and direct SNP-space residual Conv1D WGAN; also contains population-balanced sampling and PacGAN packing support.
`cwgan_extensions/conditioning.py`	Loader and validation for external AFR/EUR/AMR ancestry coefficients, including three-column ADMIXTURE-like `.Q` files.
`cwgan_extensions/selection.py`	Multi-metric loss-vector and composite-quality utilities.
`select_best_cwgan_run.py`	Compares completed runs using multiple fidelity metrics with robust candidate-set scaling.
`cwgan_extensions/privacy.py`	Exact-replay, real–synthetic proximity, and simple distance-based membership-inference diagnostics.
`evaluate_genotype_privacy.py`	Command-line wrapper for the privacy diagnostics.

1.2 Recommended ablation order

An *ablation* is an experiment in which one component is changed while the others are held fixed. A clean progression is:

1. `baseline_pca`: reproduce the present model.
2. `pca_film`: change only how ancestry information enters the network.
3. `pca_film + balanced_sampling`: then change the minibatch population mixture.
4. `pca_film + pack_size=2`: then test critic packing.
5. Repeat the best PC-space configuration with `condition_source=external`.
6. Finally test `genotype_conv`, because this changes the representation itself and therefore is not a small perturbation of the baseline.

Use the same real validation/holdout set, seed set, SNP set, and number of generated samples when comparing configurations.

2 Population-genetic notation

Suppose there are n diploid individuals and p biallelic SNPs. The genotype dosage matrix is

$$X \in \{0, 1, 2\}^{n \times p},$$

where X_{ij} is the number of alternative alleles carried by individual i at SNP j. Thus 0 means homozygous reference, 1 heterozygous, and 2 homozygous alternative.

2.1 Allele frequency and minor-allele frequency

For SNP j, the sample alternative-allele frequency is

$$\widehat{p}_j = \frac{1}{2n} \sum_{i=1}^{n} X_{ij}.$$

The minor-allele frequency (MAF) is

$$\widehat{m}_j = \min(\widehat{p}_j, 1 - \widehat{p}_j).$$

A synthetic dataset can match a PCA cloud while still having biased $\widehat{p}_j$ or $\widehat{m}_j$, so allele-frequency fidelity must be measured separately. Synthetic-genome studies commonly compare frequency distributions and other locus-level statistics [Yelmen et al., 2021, 2023, Wharrie et al., 2023].

2.2 Heterozygosity

For an individual i, the observed heterozygosity fraction over p loci is

$$H_i = \frac{1}{p} \sum_{j=1}^{p} \mathbf{1}\{X_{ij} = 1\}.$$

This quantity is especially relevant to the current reconstruction pipeline because a continuous PCA reconstruction is converted back to discrete 0/1/2 genotypes.

2.3 Linkage disequilibrium

Linkage disequilibrium (LD) means non-random association between alleles at different loci [Slatkin, 2008]. For two biallelic loci A and B on phased haplotypes, let p_A and p_B be alternative-allele frequencies and p_{AB} the frequency of the haplotype carrying the alternative allele at both loci. Then

$$D = p_{AB} - p_A p_B, \qquad r^2 = \frac{D^2}{p_A(1 - p_A)p_B(1 - p_B)}.$$

For unphased diploid dosage data, a commonly used proxy is the squared Pearson correlation between genotype dosages at the two loci. This is not identical to phase-aware haplotype LD in every setting. Therefore the current pipeline's dosage-r^2 diagnostic should be described as a genotype-correlation LD proxy, especially when the input is unphased.

2.4 Global ancestry proportions

For $K = 3$ ancestral source groups (AFR, EUR, AMR), define

$$q_i = (q_{i1}, q_{i2}, q_{i3}), \qquad q_{ik} \geq 0, \qquad \sum_{k=1}^{3} q_{ik} = 1.$$

The vector q_i lies on the two-dimensional probability simplex Δ^2. It is the condition supplied to the generator and critic.

3 Baseline WGAN-GP: what the objective means

A generator G_θ maps random latent noise $z \sim P_Z$ and ancestry condition c to a synthetic representation:

$$\tilde{x} = G_\theta(z, c).$$

The critic $C_\phi(x, c)$ returns an unrestricted real number rather than a probability. WGAN replaces a classical GAN classification objective with an integral probability metric related to Earth-mover/Wasserstein-1 distance [Arjovsky et al., 2017]. The conditional version supplies c to both networks, following the general conditional-GAN idea [Mirza and Osindero, 2014].

The critic is trained to make real samples score higher than generated samples. In the sign convention used by the code, the critic minimizes

$$L_C = \mathbb{E}[C_\phi(\tilde{x}, c)] - \mathbb{E}[C_\phi(x, c)] + \lambda_{\text{GP}} L_{\text{GP}},$$

and the generator minimizes

$$L_G = -\mathbb{E}[C_\phi(G_\theta(z, c), c)].$$

The gradient penalty is

$$L_{\text{GP}} = \mathbb{E}_{\hat{x}} \left(\|\nabla_{\hat{x}} C_\phi(\hat{x}, c)\|_2 - 1 \right)^2,$$

where $\hat{x} = \alpha x + (1 - \alpha)\tilde{x}$ and $\alpha \sim U(0, 1)$. This is the WGAN-GP construction of Gulrajani et al. [2017], used to encourage the critic to behave approximately as a 1-Lipschitz function along lines between real and generated samples.

4 Change 1: direct SNP-space generation instead of PCA reconstruction

4.1 What PCA does

Let centered real genotype vectors be rows of X_c. PCA diagonalizes the empirical covariance

$$\widehat{\Sigma} = \frac{1}{n-1} X_c^T X_c = U \Lambda U^T,$$

where the columns of U are eigenvectors (principal axes) and $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_p)$ contains descending eigenvalues. Population-genetic PCA is widely used to summarize ancestry/population structure [Patterson et al., 2006].

Retaining only the first k PCs produces the rank-k reconstruction

$$\widehat{X}_c = X_c U_k U_k^T.$$

The covariance represented by an idealized PCA reconstruction is

$$\widehat{\Sigma}_k = U_k \Lambda_k U_k^T.$$

Therefore all covariance carried exclusively by discarded components $k+1, \ldots, p$ is removed.

Proposition 1 (PCA truncation removes covariance in discarded eigendirections)**.** *Let $\Sigma = U\Lambda U^T$ and let $P_k = U_k U_k^T$. For a centered random genotype vector X with covariance Σ, the reconstruction $\widehat{X} = P_k X$ has covariance*

$$\mathrm{Cov}(\widehat{X}) = U_k \Lambda_k U_k^T.$$

Proof. Because P_k is deterministic and symmetric,

$$\mathrm{Cov}(P_k X) = P_k \,\mathrm{Cov}(X) P_k^T = P_k U \Lambda U^T P_k.$$

Using $P_k = U_k U_k^T$ and orthonormality $U_k^T U = [I_k \; 0]$ gives $U_k \Lambda_k U_k^T$. □

This does *not* imply that PCA is bad for population structure. It means that if local LD or other biologically important dependence lies partly in low-variance directions, a small k can discard it even when PC1–PC2 looks excellent.

4.2 What the independent binomial discretization does

The baseline obtains a continuous reconstructed dosage $V_j \in [0, 2]$ and then draws

$$G_j \mid V_j \sim \mathrm{Binomial}\left(2, \frac{V_j}{2}\right).$$

This is preferable to deterministic rounding for maintaining genotype variance, but it introduces locus-wise sampling noise.

Proposition 2 (Effect of independent binomial discretization on covariance)**.** *Assume that, conditional on the continuous vector V, genotype draws are independent across loci. For two distinct loci $j \neq k$,*

$$\mathrm{Cov}(G_j, G_k) = \mathrm{Cov}(V_j, V_k).$$

However,

$$\mathrm{Var}(G_j) = \mathrm{Var}(V_j) + \mathbb{E}\left[V_j\left(1 - \frac{V_j}{2}\right)\right].$$

Therefore the covariance can be retained in expectation while the marginal variances increase, which can attenuate correlation magnitudes.

Proof. By the law of total covariance,

$$\mathrm{Cov}(G_j, G_k) = \mathbb{E}[\mathrm{Cov}(G_j, G_k \mid V)] + \mathrm{Cov}(\mathbb{E}[G_j \mid V], \mathbb{E}[G_k \mid V]).$$

Conditional independence makes the first term zero. Since $\mathbb{E}[G_j \mid V] = V_j$, the second term is $\mathrm{Cov}(V_j, V_k)$. For the variance, the law of total variance gives

$$\mathrm{Var}(G_j) = \mathbb{E}[\mathrm{Var}(G_j \mid V)] + \mathrm{Var}(\mathbb{E}[G_j \mid V]),$$

and a Binomial$(2, V_j/2)$ variable has conditional variance $V_j(1 - V_j/2)$. □

The important conclusion is more nuanced than "independent sampling destroys all LD." It does not. Rather: (i) PCA truncation can already remove relevant covariance; (ii) independent discretization adds locus-specific noise and can reduce correlation coefficients; and (iii) no phase-specific haplotype pattern is explicitly modeled.

4.3 Why a 1-D convolution is a better inductive bias for ordered SNPs

A 1-D convolution with kernel width w computes local features

$$h_{t,r}^{(\ell+1)} = \sigma\left(b_r + \sum_s \sum_{u=0}^{w-1} K_{r,s,u} h_{t+u,s}^{(\ell)}\right).$$

Nearby SNPs contribute jointly to each hidden feature. Stacking convolutional layers increases the *receptive field*, the interval of input positions that can influence one hidden unit. For L stride-1 layers of width w, the receptive field is

$$R_L = 1 + L(w-1).$$

Downsampling or dilation can expand it faster.

This architecture does not assume that adjacent SNPs are independent. It can learn motifs and local correlation patterns from ordered SNPs. Large-scale artificial-genome work has specifically used convolutional WGAN-GP architectures, residual blocks, positional information, and packing to better model large genomic regions [Yelmen et al., 2023]. Earlier work also evaluated artificial genomes by allele frequency, LD, haplotypic distances, and population structure [Yelmen et al., 2021].

4.4 Residual blocks and gradient flow

A residual block computes

$$y = x + F(x;\theta).$$

Its Jacobian is

$$\frac{\partial y}{\partial x} = I + J_F(x).$$

Thus backpropagated gradients have a direct identity route in addition to the learned transformation. This does not mathematically guarantee that gradients can never vanish, but it removes the requirement that every gradient path be a product only of learned Jacobians. Residual networks were introduced to make substantially deeper networks easier to optimize [He et al., 2016].

4.5 What the code implements

Use

```
--model_backend genotype_conv
--conv_channels 128
--conv_blocks 4
--pack_size 1
```

The direct model trains on imputed dosages $0, 1, 2$, maps them to $-1, 0, 1$, and lets the generator output continuous values in $[-1, 1]$. Generated continuous values are converted to expected dosage in $[0, 2]$ and then discretized by the same Binomial$(2, p)$ mechanism.

Important limitation: genotype is not haplotype. The current CSVs are unphased diploid dosages. A row such as $(1, 1)$ at two loci does not reveal whether the chromosomes are $(0, 0) + (1, 1)$ or $(0, 1) + (1, 0)$. Therefore no model trained only on dosage can recover true phase information that is absent from the input. The direct Conv1D backend can learn local *dosage* dependencies. Claims about haplotype fidelity require phased input and phase-aware evaluation. Yelmen et al.'s large-scale convolutional experiments used phased 0/1 haplotypes [Yelmen et al., 2023].

Memory scaling. A Conv1D activation tensor has approximate storage proportional to BCL, where B is batch size, C channel width, and L SNP count. For very large L, reduce `batch_size` or `conv_channels`. The implementation prints this warning explicitly.

5 Change 2: repeated ancestry conditioning with FiLM

5.1 Why input-only concatenation can be weak

The baseline concatenates condition c to the generator's latent vector at the input, $[z, c]$, and to the critic input, $[x, c]$. This is a valid conditional architecture and, with enough capacity, can approximate rich conditional functions. The problem is not representational impossibility. The practical concern is that information from c must propagate through every subsequent transformation.

For a depth-L network $h_L = f_L \circ \cdots \circ f_1([z, c])$, the derivative of a late hidden representation with respect to c contains a product of Jacobians:

$$\frac{\partial h_L}{\partial c} = J_{f_L} J_{f_{L-1}} \cdots J_{f_1, c}.$$

If several factors contract directions associated with c, sensitivity to the condition can become small.

5.2 FiLM definition

Feature-wise Linear Modulation (FiLM) applies an affine transformation to hidden features whose parameters are functions of the condition [Perez et al., 2018]:

$$\mathrm{FiLM}(h; c) = \gamma(c) \odot h + \beta(c).$$

For the implemented MLP, both $\gamma(c)$ and $\beta(c)$ are learned affine maps. The layer is initialized near identity using $\gamma(c) = 1 + \Delta\gamma(c)$ with $\Delta\gamma = 0$ and $\beta = 0$ initially.

With FiLM inserted at multiple depths, the final output has multiple direct computational paths from c. Schematically,

$$\frac{\partial h_L}{\partial c} = \text{long path from early layers} + \sum_{\ell=1}^{L} \text{path from FiLM injection at layer } \ell.$$

This does not prove FiLM must outperform concatenation. It gives the condition repeated opportunities to modulate features rather than relying on a single input injection.

5.3 Code

Use

```
--model_backend pca_film
```

This leaves the PCA representation unchanged and is therefore the cleanest first architecture ablation.

6 Change 3: external ancestry coefficients instead of PCA-barycentric conditions

6.1 The current barycentric construction

Let $v_{AFR}, v_{EUR}, v_{AMR} \in \mathbb{R}^k$ be reference-population centroids in PC space. The current condition for a query sample with PC coordinate y approximately solves

$$y \approx q_{AFR} v_{AFR} + q_{EUR} v_{EUR} + q_{AMR} v_{AMR}, \qquad \sum_k q_k = 1,$$

by least squares, followed by clipping negative coefficients and renormalization.

This is geometrically intuitive, but it makes the condition a deterministic function of the same PC representation that the baseline generator is trained to generate. Consequently, success at generating PC coordinates consistent with c is partly built into how c was defined. This is not "data leakage" in the machine-learning train/test sense; it is a representational circularity that can make conditional interpretation less independent.

6.2 ADMIXTURE-style model-based ancestry

ADMIXTURE estimates global ancestry using a generative allele-frequency model rather than defining ancestry as coordinates inside the PCA space [Alexander et al., 2009]. A simplified formulation is

$$X_{ij} \sim \text{Binomial}(2, \pi_{ij}), \qquad \pi_{ij} = \sum_{k=1}^{K} q_{ik} f_{kj},$$

where q_{ik} is individual i's ancestry fraction from source k and f_{kj} is the alternative-allele frequency of SNP j in source k.

PCA and ADMIXTURE answer related but different questions: PCA provides axes summarizing covariance/population structure, while model-based ancestry methods directly estimate mixture coefficients under a specified latent-population model. Alexander et al. explicitly contrast PCA-based approaches with model-based ancestry estimation [Alexander et al., 2009].

Important nuance. If ADMIXTURE and PCA are computed from the same genotype matrix, the estimates are not statistically independent data. The advantage is that the condition is no longer defined as a direct barycentric transformation of the model's output coordinates.

6.3 Code

The pipeline accepts either CSV/TSV files with named columns `AFR,EUR,AMR` or a headerless three-column ADMIXTURE-like `.Q` file:

```
--condition_source external \
--condition_input BOG=data/BOG/BOG.Q \
--external_condition_order AFR,EUR,AMR
```

For combined models, pass one `-condition_input NAME=PATH` per constituent dataset. Row order must exactly match the genotype CSV row order. The loader checks row counts, non-negativity, finite values, and normalizes every row to the simplex.

7 Change 4: population-balanced minibatch sampling

7.1 The imbalance problem

Suppose the training set contains groups $g = 1, \ldots, G$ with sizes n_g. Uniformly sampling individuals optimizes an empirical objective of the form

$$L_{\text{emp}}(\theta) = \sum_{g=1}^{G} \frac{n_g}{N} \left(\frac{1}{n_g} \sum_{i \in g} \ell_i(\theta) \right), \qquad N = \sum_g n_g.$$

Large groups receive larger coefficients n_g/N. This is appropriate if the desired training distribution is exactly the observed pooled sample mixture. It is not appropriate if the goal is to give a small admixed query cohort comparable influence to a much larger reference set.

7.2 Inverse-frequency weighted sampling

Assign individual i in group $g(i)$ sampling weight

$$w_i \propto \frac{1}{n_{g(i)}}.$$

Then the total sampling mass of each group is

$$\sum_{i \in g} w_i \propto n_g \frac{1}{n_g} = 1.$$

After normalization, each group has probability $1/G$. Therefore the expected stochastic gradient corresponds to the group-balanced objective

$$L_{\text{bal}}(\theta) = \frac{1}{G} \sum_{g=1}^{G} \left(\frac{1}{n_g} \sum_{i \in g} \ell_i(\theta) \right).$$

This is the precise sense in which the sampler "balances" the data. It changes the optimization target; it does not create additional biological information.

7.3 Code and groups

Use

```
--balanced_sampling
```

with `pca_film` or `genotype_conv`. Query groups are labeled separately (for example `QUERY_BOG`, `QUERY_BAQ`); reference groups are `REF_AFR`, `REF_EUR`, `REF_AMR`, and `REF_OTHER` if present. The run log prints observed group counts.

When not to use it. If the scientific target is the empirical pooled population distribution, equalizing groups biases the training objective away from that distribution. Treat balanced sampling as a hypothesis to test, not an automatic correction.

8 Change 5: PacGAN-style critic packing to probe mode collapse

8.1 Mode collapse

Let P be the real distribution and Q_θ the generator distribution. *Mode collapse* means Q_θ places too little or zero probability on regions that have appreciable probability under P, even if generated samples individually look plausible.

8.2 Packing

PacGAN changes the critic from a function of one sample to a function of a pack of m samples [Lin et al., 2018]:

$$C(x_1, \ldots, x_m, c_1, \ldots, c_m).$$

The real and generated distributions seen by the critic become product distributions P^m and Q^m.

A simple calculation shows why missing modes can become easier to detect. Suppose a region A has real mass $P(A) = \varepsilon > 0$ but the collapsed generator has $Q(A) = 0$. In one sample, the probability of seeing the missing region under P is ε. In a pack of m independent real samples, the probability that at least one lies in A is

$$1 - (1 - \varepsilon)^m,$$

whereas the generator's probability remains 0. Since $1 - (1 - \varepsilon)^m$ increases with m, the missing mode creates a stronger pack-level signal.

8.3 Code

Use, for example,

```
--pack_size 2
```

with `pca_film` or `genotype_conv`. In vector space, packed samples are concatenated along the feature dimension. In Conv1D space, packed samples become input channels to the critic. Conditions are packed consistently with their samples. The gradient penalty is then computed with respect to the packed critic input.

Packing reduces the number of independent critic decisions per minibatch by a factor of m. Very large m can therefore increase variance or memory use; $m = 2$ is a reasonable first ablation.

9 Change 6: multi-metric model selection instead of selecting by GAN loss alone

9.1 Why adversarial loss is insufficient

WGAN losses are useful training diagnostics [Arjovsky et al., 2017], but a low adversarial objective does not logically imply correct MAF, LD, heterozygosity, population structure, or privacy. Genomic synthetic-data studies evaluate multiple properties because no single statistic captures all biologically relevant structure [Yelmen et al., 2021, 2023, Wharrie et al., 2023].

The code already records MAF error, heterozygosity, dosage-LD error, RBF-MMD in PC space, and nearest-neighbor adversarial accuracy. MMD is a distributional two-sample discrepancy in a reproducing-kernel Hilbert space [Gretton et al., 2012].

9.2 Loss orientation

The selector converts all metrics to "smaller is better" losses:

$$\begin{aligned}
\ell_{MAF} &= \mathrm{MAE}(MAF_{real}, MAF_{syn}),\\
\ell_{H} &= |\overline{H}_{real} - \overline{H}_{syn}|,\\
\ell_{LD} &= \mathrm{MAE}(r^2_{real}, r^2_{syn}),\\
\ell_{MMD} &= \mathrm{MMD}^2,\\
\ell_{AA} &= |AA - 0.5|.
\end{aligned}$$

For the nearest-neighbor adversarial-accuracy statistic, 0.5 corresponds to chance-level local distinguishability, hence the absolute distance from 0.5.

9.3 Why scaling is necessary

These metrics have different numerical units and ranges. Adding raw values would make the largest-scale metric dominate arbitrarily. Across candidate models, the selection script estimates a robust scale s_j for each metric using its interquartile range (falling back to median absolute deviation when needed), then computes

$$S_i = \frac{\sum_j w_j \ell_{ij}/s_j}{\sum_j w_j}.$$

Lower S_i is better under the chosen weights.

This score is *candidate-set dependent*; it is not a universal biological-quality number. It should compare models evaluated on the same SNPs and the same independent holdout cohort.

9.4 Code

```
python select_best_cwgan_run.py 'runs/20261003_*' \
    --output model_comparison.csv
```

Weights can be overridden with repeated `-weight METRIC=VALUE`. The shared trainer in `cwgan_extensions/models.py` also exposes an `epoch_callback` interface that can restore the best epoch under an externally supplied selection score. It is deliberately not enabled by default because a checkpoint score requires a fixed validation protocol and explicit metric scaling choices.

10 Change 7: privacy evaluation separated from fidelity evaluation

10.1 Fidelity and privacy are different questions

A model that exactly memorizes the training set can have excellent MAF, LD, and PCA fidelity. Therefore fidelity cannot prove privacy. Recent genomic synthetic-data evaluations include relatedness, nearest-neighbor, membership-inference, and multi-representation leakage analyses [Yelmen et al., 2023, Wharrie et al., 2023, Correa Rojo et al., 2026].

10.2 Exact replay

An exact replay occurs if a synthetic genotype row is byte-for-byte identical to a real genotype row. The code reports the fraction of training and holdout real rows that appear exactly in the synthetic set. Zero exact replay is reassuring but extremely weak: a nearly copied genome is not an exact replay.

10.3 Distance-based membership inference

Let $\phi(x)$ be a standardized PCA embedding and let S be the synthetic set. Define

$$d(x, S) = \min_{y \in S} \|\phi(x) - \phi(y)\|_2^2, \qquad s(x) = -d(x, S).$$

If training members are systematically closer to the synthetic set than real non-members held out from training, $s(x)$ can reveal membership. The script reports ROC AUC for distinguishing training members from holdout non-members. AUC $= 0.5$ means this particular attack is at chance; it does not establish privacy.

10.4 Differential privacy is a stronger formal guarantee

A randomized training algorithm $\mathcal{A}$ is (ε, δ)-differentially private if, for every pair of neighboring datasets D, D' differing in one individual and every measurable output set S,

$$\Pr[\mathcal{A}(D) \in S] \leq e^{\varepsilon} \Pr[\mathcal{A}(D') \in S] + \delta.$$

None of the diagnostics implemented here proves this inequality. They are empirical audits for memorization/proximity leakage only.

10.5 Code

Privacy evaluation requires a real training set and a genuinely unused real holdout set:

```
python evaluate_genotype_privacy.py \
  --train_real real_train.csv \
  --holdout_real real_holdout.csv \
  --synthetic runs/.../outputs/synthetic_genotypes.csv \
  --output privacy_metrics.json
```

Do not use the same individuals in `train_real` and `holdout_real`.

11 Relationship between the changes

The changes target different failure modes and should not be treated as substitutes.

Change	Failure mode targeted	What improvement would look like
Direct Conv1D SNP model	PCA rank truncation; weak explicit local-locus inductive bias	Better LD/MAF and higher-order local dependence without loss of ancestry structure
FiLM conditioning	Condition influence may weaken through depth	Better ancestry-conditional fidelity at matched unconditional quality

External ancestry q	Condition defined directly from output PC geometry	Conditional behavior remains good under independently modeled ancestry coefficients
Balanced sampling	Large groups dominate SGD expectation	Better target-query fidelity without unacceptable loss on reference structure
Packing	Generator covers too few modes	Better diversity/coverage and adversarial accuracy without frequency/LD degradation
Multi-metric selection	GAN loss does not encode biological fidelity	Selected runs improve several hold-out metrics simultaneously
Privacy audit	High-fidelity memorization can go unnoticed	Low replay/proximity/membership signal while retaining utility

12 Concrete command-line experiments

12.1 Baseline reproduction

```
python conditional_wgan_pipeline.py \
  --datasets BOG \
  --dataset_input BOG=data/BOG/genotypes.csv \
  --ref_input data/reference/reference.csv \
  --fam data/reference/reference.fam \
  --model_backend baseline_pca \
  --val_fraction 0
```

Setting `val_fraction=0` restores use of all samples for training and disables a real validation split. The original code did not use training data "as validation"; it simply had no validation set.

12.2 FiLM only

```
python conditional_wgan_pipeline.py ... \
  --model_backend pca_film \
  --pack_size 1
```

12.3 FiLM plus balanced sampling

```
python conditional_wgan_pipeline.py ... \
  --model_backend pca_film \
  --balanced_sampling
```

12.4 FiLM plus PacGAN packing

```
python conditional_wgan_pipeline.py ... \
  --model_backend pca_film \
  --pack_size 2
```

12.5 External ancestry coefficients

```
python conditional_wgan_pipeline.py ... \
  --model_backend pca_film \
  --condition_source external \
  --condition_input BOG=data/BOG/BOG.Q \
  --external_condition_order AFR,EUR,AMR
```

For a combined BOG+BAQ model, both condition files are required:

```
--datasets '{BOG,BAQ}' \
--dataset_input BOG=data/BOG/genotypes.csv \
--dataset_input BAQ=data/BAQ/genotypes.csv \
--condition_input BOG=data/BOG/BOG.Q \
--condition_input BAQ=data/BAQ/BAQ.Q \
--condition_source external \
--bog_baq_harmonized
```

12.6 Direct SNP-space residual convolutional model

Start conservatively because memory use is much larger than for 50 PCs:

```
python conditional_wgan_pipeline.py ... \
  --model_backend genotype_conv \
  --batch_size 8 \
  --conv_channels 64 \
  --conv_blocks 4 \
  --pack_size 1
```

Then test `-pack_size 2` and/or `-balanced_sampling` as separate ablations.

13 How to interpret outcomes

13.1 Do not optimize only PC overlap

PC1–PC2 can look convincing while locus-level or local-dependence statistics are wrong. Conversely, a direct SNP model can improve local statistics while slightly changing the PCA cloud. HAPNEST's evaluation explicitly treats MAF, population structure, LD, and nearest-neighbor adversarial accuracy as separate fidelity properties and reports trade-offs among metrics [Wharrie et al., 2023].

13.2 Recommended minimum acceptance table

For every run, compare at least:

a) MAF MAE and correlation;

b) heterozygosity mean and distribution;

c) LD/dosage-r^2 error versus physical distance if genomic positions are available;

d) PC-space MMD and PC alignment;

e) nearest-neighbor adversarial accuracy;

f) exact replay and membership-inference AUC using an independent holdout set.

The existing adjacent-SNP LD metric is useful for smoke tests but should be upgraded to distance-binned LD decay on non-LD-pruned SNPs for serious validation. LD-pruned training data intentionally removes much of the local correlation one would otherwise ask the model to reproduce.

14 What is proven here and what is empirical

It is important to separate mathematical facts from empirical hypotheses.

- **Proven/algebraic:** rank-k PCA reconstruction has covariance $U_k \Lambda_k U_k^T$; inverse-frequency sampling equalizes group sampling mass; conditional-independent binomial discretization preserves off-diagonal covariance in expectation but adds marginal variance; a residual block has Jacobian $I + J_F$; a missing mode appears in a real pack with probability $1 - (1 - \varepsilon)^m$.
- **Architectural rationale, not a theorem of better performance:** Conv1D, FiLM, residual blocks, and packing provide inductive biases or optimization paths that may help this dataset. They must be tested empirically.
- **Not a privacy guarantee:** low membership AUC, low nearest-neighbor leakage, or zero exact replay do not imply differential privacy.

15 Suggested experiment log

For each comparison, record:

1. exact git/file version or checksum of the code;
2. genotype and reference input hashes;
3. whether BOG/BAQ harmonization was completed and how alleles/strand were checked;
4. whether data were LD-pruned and, if so, the pruning threshold/window;
5. ancestry condition source and, for external conditions, ADMIXTURE K, reference panel, and column mapping;
6. backend, seed, validation fraction, balanced-sampling flag, pack size, and all model hyperparameters;
7. fidelity metrics on a fixed holdout set;
8. privacy diagnostics on train versus independent holdout individuals.

This is more valuable than selecting a single visually pleasing PCA plot because it makes results reproducible and scientifically interpretable.

16 Implementation status and limitations

The supplied code was smoke-tested on small synthetic CSV inputs for all three backends: `baseline_pca`, `pca_film` with balancing/packing, and `genotype_conv`. The external-condition loader, multi-run selector, and privacy command-line tool were also executed end-to-end on small test data.

The direct Conv1D backend is an experimental implementation inspired by the genomic convolutional literature, not a line-for-line reproduction of Yelmen et al. [Yelmen et al., 2023]. In particular, it operates on unphased diploid dosage rather than phased 0/1 haplotypes, uses a simple learned locus-specific output bias plus a positional critic channel, and currently processes the full SNP vector rather than an explicitly windowed chromosome hierarchy. These differences should be stated whenever results are reported.

17 Summary

The most important conceptual change is to separate three distinct goals:

1. **Representation fidelity**: does the model preserve allele frequencies, genotype correlations/LD, heterozygosity, population structure, and higher-order dependence?
2. **Conditional fidelity**: does changing $c = (q_{AFR}, q_{EUR}, q_{AMR})$ change generated genotypes in the intended ancestry-consistent way?
3. **Privacy/generalization**: does the model generate new individuals rather than reproducing the training genomes?

No single GAN loss or PCA figure answers all three. The modular code is designed so each proposed mechanism can be isolated, measured, and rejected if it does not improve the relevant holdout metrics.

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